The study of Immune microenvironment of Korean NSCLC patients

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Introduction

According to the findings of a survey from Institute for Health Metrics and Evaluation (IHME), cancer is the second cause of death worldwide[1]. In 2019, Lung & Bronchus cancer was estimated a second most common new cancer cases through both sex, and most death cancer cases in both males and females[2]. About 80 percent to 85 percent of lung cancers are non-small cell lung cancer(NSCLC)[3]. Lung adenocarcinoma(LUAD) is the most common type of lung cancer seen in non-smokers, and smokers[3]. Therefore, understanding genetic and epigenetic interaction between genes are important to understand lung cancer. We conducted research on NSCLC and the immune landscape. Our co-researcher team divided the subtypes of the samples by using the NMF(Non-negative Matrix Factorization) clustering method using global protein, phospho and acetyl data. We attempted to explain this sample subtype with different methodologies for the gene or sample.

Method

Immune clustering

We used RNA expression data of 205 Tumor samples and 85 Normal adjacent tissues (NATS) samples generated from Asan Medical Center. We verified NMF subtypes(4 clusters) in each method using RNA expression data as described in the refs[6,7,8,9]. In the Immunogram clustering method, GSVA was performed with gene sets involved in the Cancer-immunity cycle(CLC), followed by K-means clustering(Hot, Cold)[6]. In other method, samples was clustered by Consensus Clustering(Hot, Cold, NAT_enriched) using xcell scores converted from RNA expression data[7,8]. Finally, in the Fges clustering method, ssGSEA was performed for 28 gene sets related to immune invasion to calculate nes for each sample. After that, our data was clustered by Louvain clustering(Immune-enriched, fibrotic(IE/F), Immune-enriched, non-fibrotic(IE), fibrotic(F), immune-depleted(D))[9].
Among the various methods, the xCell and Immunogram methods were distinguished the NMF clusters well, so our results were summarized based on the xCell and Immunogram methods.

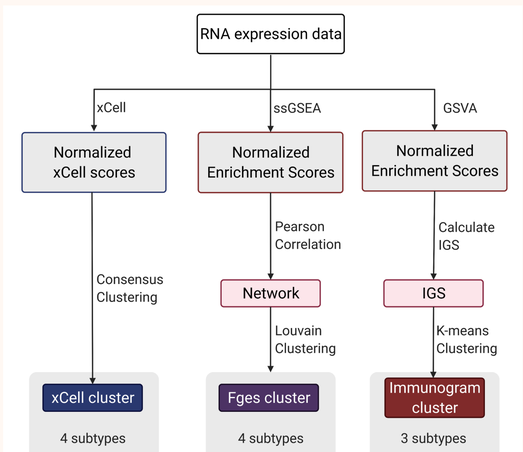

Fig1. Overview of Input data

Result

Immune clustering result

Hot-tumor-enriched (HTE) and cold-tumor-enriched (CTE) immune clusters were considerably overlapped between cell type- (xCell) and pathway-based (Immunogram) clustering. HTE tumors were enriched of B cells, CD8+ and CD4+ T cells, dendritic cells, macrophages, and regulatory T cells in cell type level when they were compared to CTE tumors. In the pathway level clustering, HTE cluster showed higher activation in immune-related pathways than CTE. Both characteristics of HTE and CTE were reported in the previous studies of immune profiling of lung cancer patients[7,8]. Each enrichment score of cell types and pathways along with immune cluster were then tested against patient survival.

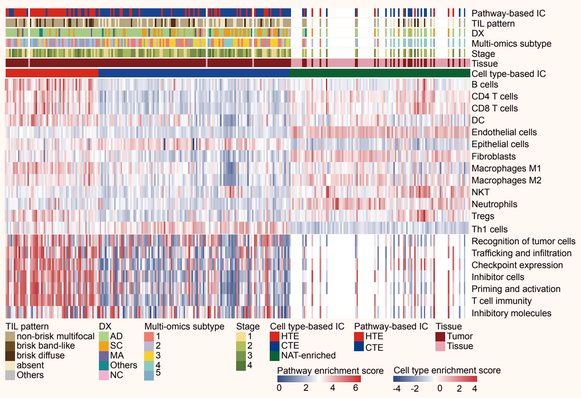

Fig2. Immune clustering results

Survival analysis result

The immune cluster was the most favorable factor for patient survival and high enrichment of CD8+ and CD4+ T cells, B cells, natural killer cells, and basophils showed a positive correlation with good prognosis in contrast to epithelial cells and neutrophils having negative correlation. The pathway-based enrichment scores of the patients could not explain the survival statistically although most of them showed a positive correlation with longer survival.

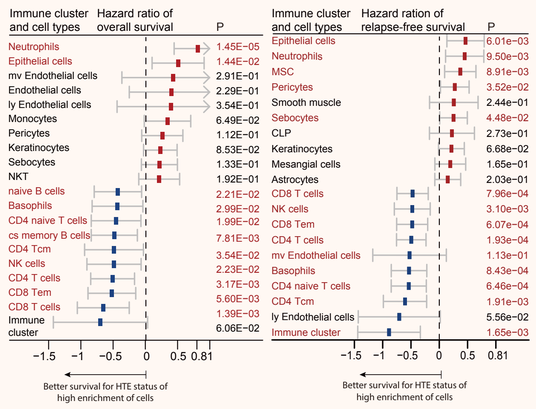

Fig3. The hazard ratios of immune and stroma cell

Pathway analysis enriched in the immune cluster at the multi-omics level

A set of immune-related pathways and EMT signaling were enriched in HTE tumors in multi-omics level. In the other hand, cell cycle-related pathways and glycolysis were enriched in CTE in multi-omics level. Specifically, some of metabolic pathways and oxidative phosphorylation were enriched in CTE tumors in phospho-protein level but they were enriched in HTE tumors in acetyl-protein level. Many of our pathway results were concordant with the findings of the previous lung cancer multi-omics studies[7,8].

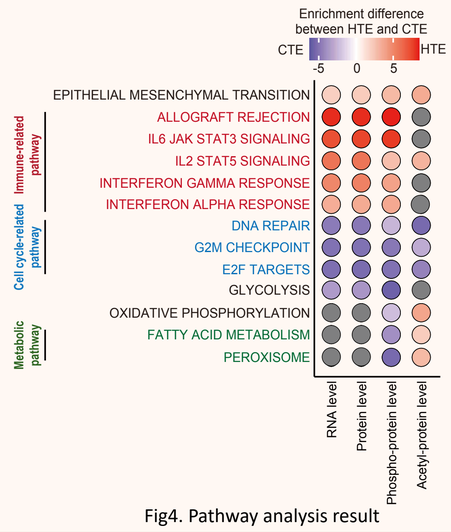

Fig4. Pathway analysis result

Future work

Our plan is to predict clinical traits(i.e., overall-survival, metastasis) with multi-omics data. Finding biomarkers that outstanding describes the characteristics of samples may contribute to a progressing understanding of the correlation between alterations at the multi-omics level and the patients' clinical traits. The next goal is to find a biomarker that can predict the prognosis for each treatment.

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